

Introduction to Mathematics and Optimisation

Exercises Week 2

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1 Exercise 1.89

Prove by induction:

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

1.1 Base Case (n=1)

We test the formula for $n = 1$.

$$\begin{aligned} \sum_{i=1}^1 i^3 &= 1^3 = 1 \\ \left(\frac{1(1+1)}{2} \right)^2 &= \left(\frac{1 \cdot 2}{2} \right)^2 = 1^2 = 1 \end{aligned}$$

The formula holds true for $n = 1$.

1.2 Inductive Hypothesis

Assume formula is true for some $n = k$

$$\sum_{i=1}^k i^3 = \left(\frac{k(k+1)}{2} \right)^2$$

1.3 Inductive Step

We must now prove that the formula is true for $n = k + 1$.

$$\sum_{i=1}^{k+1} i^3 = \left(\frac{(k+1)((k+1)+1)}{2} \right)^2$$

$$\begin{aligned}
\sum_{i=1}^{k+1} i^3 &= \sum_{i=1}^k i^3 + (k+1)^3 \\
&= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3 \\
&= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\
&= \frac{k^2(k+1)^2}{4} + \frac{4(k+1)^3}{4} \\
&= \frac{(k+1)^2(k^2 + 4(k+1))}{4} \\
&= \frac{(k+1)^2(k^2 + 4k + 4)}{4} \\
&= \frac{(k+1)^2(k+2)^2}{4} \\
&= \left(\frac{(k+1)(k+2)}{2} \right)^2 \\
&= \left(\frac{(k+1)((k+1)+1)}{2} \right)^2
\end{aligned}$$

1.4 Simplified LEAN proof

Here is proof using LEAN constructed by me

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import Mathlib
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theorem sum_of_first_n_cubes (n : ℕ) : ∑ i ∈ Finset.range (n + 1), i^3 =
  ((n * (n+1)) / 2)^2 := by
  have four_mul_sum_eq : 4 * ∑ i ∈ Finset.range (n + 1), i^3 =
    (n * (n+1))^2 := by
    induction n with
    | zero =>
      simp
    | succ k ih =>
      rw [Finset.sum_range_succ]
      calc 4*(∑ i ∈ Finset.range (k + 1), i^3 + (k + 1)^3) =
        4 * ∑ i ∈ Finset.range (k + 1), i^3 + 4*(k + 1)^3 := by
          rw [mul_add]
        _ = (k * (k + 1))^2 + 4 * (k + 1)^3 := by rw [ih]
      ring
  have mul_div_cancel : (4 * ∑ i ∈ Finset.range (n + 1), i^3) / 4 =
    ∑ i ∈ Finset.range (n + 1), i^3 := by
    rw [Nat.mul_div_cancel_left]

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decide
have div_pow_eq : ((n * (n + 1))^2) / 4 = ((n * (n + 1)) / 2)^2 := by
  obtain ⟨k, hk⟩ := Nat.even_mul_succ_self n
  have h : 2 ∣ n * (n + 1) := ⟨k, by rw [two_mul]; exact hk⟩
  rw [Nat.div_pow h]
rw [← div_pow_eq]
calc
  ∑ i ∈ Finset.range (n + 1), i^3 =
    (4 * ∑ i ∈ Finset.range (n + 1), i^3) / 4 := by
      rw [mul_div_cancel]
  _ = (n * (n + 1))^2 / 4 := by rw [four_mul_sum_eq]

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2 Exercise 1.114

What is $f^{-1}(C)$ when $A = \mathbb{R}$, $B = \mathbb{R}$, $f(x) = x^2 - 5x + 6$ and $C = (-\infty, 0]$?

2.1 Solution

We need to find a set of all x values that map to the interval C . Since $C = (-\infty, 0]$ we can define the preimage of f under C as:

$$f(x) \leq 0$$

To solve this we need to solve the quadratic inequality

$$x^2 - 5x + 6 \leq 0$$

To solve the quadratic inequality we need to solve the following quadratic equation

$$\begin{aligned}
 x^2 - 5x + 6 &= 0 \\
 (x - 3)(x - 2) &= 0 \\
 x_1 &= 3 \\
 x_2 &= 2
 \end{aligned}$$

Since the function gives convex parabola the inequality will hold true in the interval between 2 and 3, with 2 and 3 included as the C interval includes 0.

This leaves us with the final interval of

$$[2, 3]$$

3 Exercise 1.117

Consider the three perceptrons $p_1, p_2, p_3 : \mathbb{R}^2 \rightarrow \mathbb{R}$, where

$$p_1(x, y) = \begin{cases} 1 & \text{if } -x - y > -\frac{3}{2}, \\ 0 & \text{if } -x - y \leq -\frac{3}{2}, \end{cases} \quad p_2(x, y) = \begin{cases} 1 & \text{if } x + y > \frac{1}{2}, \\ 0 & \text{if } x + y \leq \frac{1}{2}, \end{cases}$$

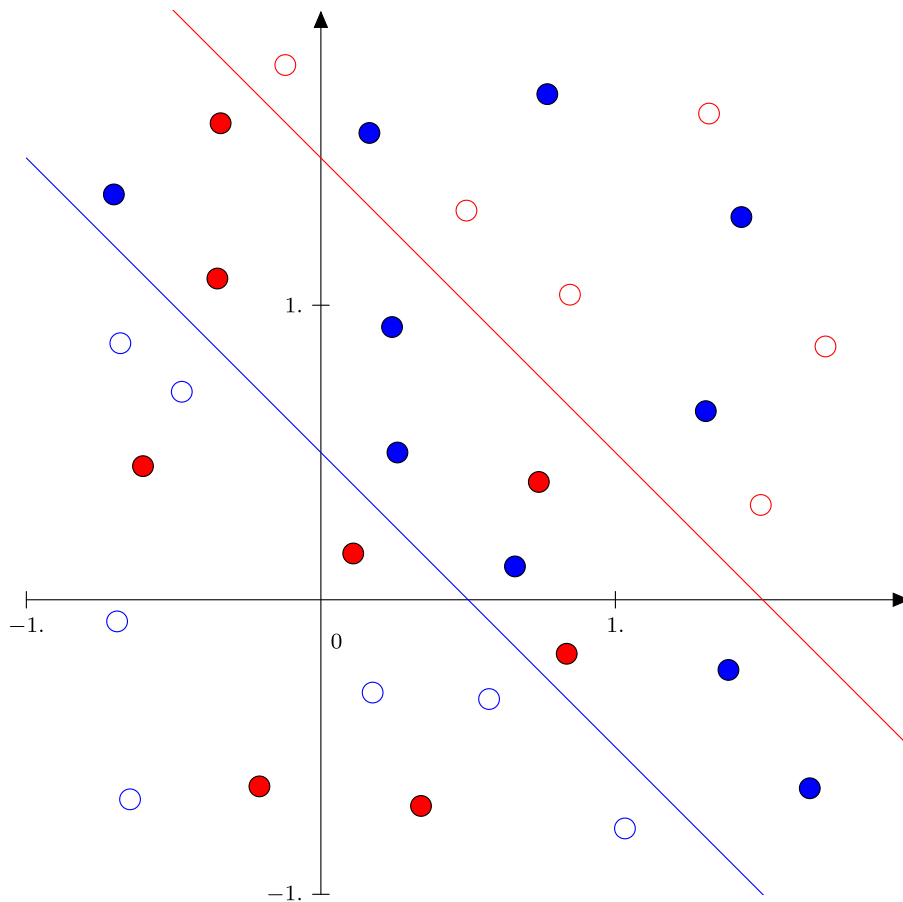
and

$$p_3(x, y) = \begin{cases} 1 & \text{if } x + y > \frac{3}{2}, \\ 0 & \text{if } x + y \leq \frac{3}{2}. \end{cases}$$

Let $f(x, y) = p_3(p_1(x, y), p_2(x, y))$. Then f is a composite function $f = g \circ h$ of two functions $h : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$. Write down these functions.

Compute $f(0, 0)$, $f(1, 0)$, $f(0, 1)$ and $f(1, 1)$.

Relate the perceptrons p_1 and p_2 to the illustration below. What do you think the red and blue line illustrate? What does it mean that a dot is solid compared to hollow? What is special about points between the red and blue lines? Try to relate $f(0, 0)$, $f(1, 0)$, $f(0, 1)$ and $f(1, 1)$ to the illustration.



(Illustration courtesy of William Heyman Krill).

3.1 Functions h and g

- $h(x, y) = (p_1(x, y), p_2(x, y))$
- $g(a, b) = p_3(a, b) = \begin{cases} 1 & \text{if } a + b > \frac{3}{2} \\ 0 & \text{if } a + b \leq \frac{3}{2} \end{cases}$

3.2 Calculations

Calculating the values we receive the truth table for logical *xor* which makes sense since p_1 produces truth table for *nand*, p_2 for *or* and p_3 for *and* and $\neg(A \wedge B) \wedge (A \vee B)$ is one of the ways to produce $A \oplus B$

- $f(0, 0) = 0$
- $f(1, 0) = 1$
- $f(0, 1) = 1$
- $f(1, 1) = 0$

3.3 Image

The inequity producing 1 in p_1 can be rewritten as:

$$\begin{aligned} -x - y &> -\frac{3}{2} \\ -y &> x - \frac{3}{2} \\ y &< -x + \frac{3}{2} \end{aligned}$$

This inequity defines the plane below the red line any point below the red (the red filled ones) produce 1 in the p_1 perceptron.

The inequity producing 1 in p_2 can be rewritten as:

$$\begin{aligned} x + y &> \frac{1}{2} \\ y &> -x + \frac{1}{2} \end{aligned}$$

This means any point above the blue line (the blue filled points) will produce 1 in the p_2 perceptron.

Any point between those two lines produces 1 in both p_1 and p_2 perceptron.